

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

**Course: B. Tech. Branch: Electronics Engineering/ Electronics and
Communication Engineering / Electronics and Telecommunication**

Subject Code & Name: BTEXPE702D/BTETPE702D & Fiber Optic Communication

Semester: VII

Max Marks: 60

Date:04-01-24

Duration: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define the Refractive index and Explain Snell's law for refraction and reflection.	Level 1	6
B) Explain the following terms: (a) Numerical Aperture (b) Total Internal Reflection (c) Acceptance angle	Level 2	6
C) Calculate the numerical aperture, fractional refractive index Δ and acceptance angle of a fiber having the refractive index of core is 1.44 and the refractive index of cladding is 1.40.	Level 2	6
Q.2 Solve Any Two of the following.		12
A) Define and explain Optical Time Domain Reflectometer.	Level 1	6
B) Define mode and explain multimode fiber in details.	Level 1	6
C) A Multimode step index fiber with a core diameter of 80 μm and a relative index difference of 1.5% is operating at a wavelength of .85 μm . If the core refractive index is 1.48, estimate (a) normalized frequency of the fiber; (b) the number of guided modes.	Level 1	6
Q. 3 Solve Any Two of the following.		12
A) Define LED and explain the operation of LED.	Level 1	6
B) Explain the working principles of PIN-diodes.	Level 2	6
C) Explain the role of optical receivers in an optical communication system.	Level 2	6
Q.4 Solve Any Two of the following.		12
A) Explain the operating principles of Erbium-Doped Fiber Amplifiers (EDFAs) as optical amplifiers.	Level 1	6
B) Explain the fundamental principles of Wavelength Division Multiplexing (WDM) networks.	Level 2	6
C) Explain the working principles of electro-optic switches and how they are used to control the flow of optical signals.	Level 2	6

Q. 5 Solve Any Two of the following.

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| A) Define self-phase modulation and elaborate on how it occurs in optical fibers. | Level 1 | 6 |
| B) What is the principle behind soliton-based communication in optical fibers? | Level 1 | 6 |
| C) Explain how solitons are formed and maintained, and why they are significant for long-distance communication. | Level 1 | 6 |

***** End *****